QUICK START & SAFETY GUIDE · SKU PIA24023P

From the Box to Live EEG over Bluetooth

Six steps: connect and charge the battery, attach the electrodes, switch on, connect over Bluetooth from a browser, phone or computer, and check the signal.



WARNING Battery only while worn.

IronBCI-8 runs from its LiPo battery and sends data wirelessly. **Never wear it while charging** — unplug the USB cable before electrodes go on. IronBCI-8 is **not a medical device**.

01 SAFETY RULES — READ BEFORE PUTTING ELECTRODES ON ANYONE

#	Rule
1	**Battery only while worn.** Nothing is cabled between the person and a computer.
2	**Never wear it while charging.** Unplug the USB cable before electrodes go on. While charging, set Switch 2 OFF.
3	**No mains connection of any kind** while electrodes are on a person.
4	**No diagnosis.** Do not use IronBCI-8 to diagnose, monitor or treat any medical condition.
5	**Implanted devices.** Not on a person with a pacemaker or other implant without a physician's approval.
6	**Skin.** Clean, intact skin only. Stop if it becomes irritated; clean off gel after each session.
7	**Exposed electronics.** No case: handle by the edges, keep liquids away.

Power while recording

Setup	While electrodes are on	Why
LiPo, both switches ON, USB unplugged	SAFE	No path to the mains.
Data received over Bluetooth	SAFE	No cable to the person.
USB cable plugged in for charging	NEVER	Connects the person to the mains through the charger.

LiPo battery care

- **Check the polarity** of every new battery before connecting it.
- Charge only through the board's Micro USB port from a 5 V supply; do not leave it charging unattended.
- Do not use a battery that is swollen, punctured, hot or damaged; recycle it at a battery collection point.
- Do not short the wires; keep the battery away from heat and water. For storage, set both switches OFF.

02 WHAT YOU NEED

Item	Requirement
IronBCI-8	Included
LiPo battery	1S, 200 mAh recommended, 2.0 mm JST, polarity checked
Micro USB cable + 5 V charger	For charging only
Electrodes	8 dry or wet Ag/AgCl with 2.54 mm connectors + 2 ear clips
Cap or headset	10-20 cap, or the open 3D-printable IronBCI headset
Receiver	Chrome / Edge (PiEEG Cloud), Android phone, or a Windows / macOS / Linux computer with Bluetooth LE
Gel	Only for wet electrodes

03 SETUP IN SIX STEPS

1 Connect the battery

The JST connector is on the underside of the stack. Match the black wire to **GND**, then plug in.

2 Charge

Set **Switch 1 ON, Switch 2 OFF** and connect Micro USB to a 5 V charger. **LED 1** lights while charging, **LED 2** when full.

3 Attach the electrodes

Channels 1–8 to the cap electrodes, **Ref** to one earlobe, **Biasout** to the other (IB8-PIN-004). Keep leads short and the device close to the head.

4 Switch on

Unplug the USB cable and set **both switches ON**. IronBCI-8 starts advertising as **EAREEG** (some units **IronBCI**). No pairing needed.

5 Connect

Pick one:

- **PiEEG Cloud**: open cloud.pieeg.com in Chrome or Edge → PiEEG Devices → IronBCI → choose the device.
- **Android**: install mobile_sdk.apk, set **8 ch** in settings, connect.
- **Computer**: `pip install "pieeg-server[ironbci]"` then `pieeg-server --device ironbci8` and open http://localhost:1617.

6 Check the signal

The **BLE LED** turns on and 8 channels stream at 250 samples per second. Turn on a 1–40 Hz band-pass filter.

04 FIRST SIGNAL CHECKS

Test	What you should see
Sit still for 30 s	All 8 channels settle. Flat lines or huge values mean a loose electrode.
Close the eyes for 10 s	Alpha (8–13 Hz) rises, strongest on Pz, then drops when the eyes open.
Blink a few times	Large slow spikes on the front channels (Fz, F7, F8).
Clench the jaw	Bursts of high-frequency noise — muscle (EMG), not brain activity.

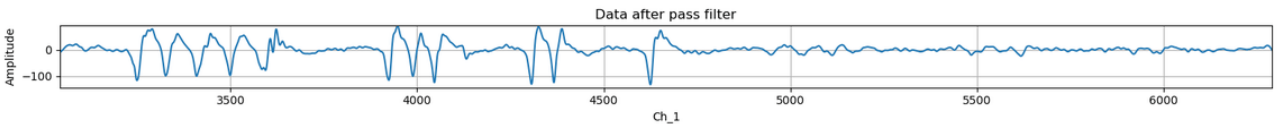


Figure 1 — Chewing and blinking artifacts in raw data: learn to recognise them before any analysis.

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05 TROUBLESHOOTING

Battery and power

Problem	Fix
Nothing happens when switched on	Check the battery is charged and plugged in, and both switches are ON.
Board warm or dead after a new battery	Polarity may be reversed. Disconnect at once and check the wires against GND.
LED 1 does not light when charging	Switch 1 ON, Switch 2 OFF. Try another cable and a 5 V charger.
Battery runs out quickly	Charge until LED 2 lights. Replace old or swollen cells.

Bluetooth connection

Problem	Fix
Device not found	Both switches ON; look for EAREEG or IronBCI; move closer.
Found but will not connect	Already connected to another app or phone. Close it, switch off and on, retry.
Connected, but no data	Switch 2 must be ON. Disconnect, switch off and on, reconnect.
Connection drops	Stay within a few metres, line of sight; avoid crowded 2.4 GHz areas.
No Bluetooth button in PiEEG Cloud	Use Chrome or Edge over HTTPS. Safari, Firefox and iOS browsers lack Web Bluetooth.
PiEEG Server cannot find it	`--ble-name IronBCI` or `--ble-address` from 1.Find_Device.py; install the [ironbci] extra.
macOS / Linux: no devices	macOS: allow Bluetooth for the terminal. Linux: sudo systemctl start bluetooth.

Signal

Problem	Fix
All channels noisy at once	Check the **REF and BIAS ear clips** first.
One channel flat or very large	That electrode is not touching the skin.
Values in the thousands of µV	Normal DC offset in raw data. Turn on a band-pass filter.
Strong 50/60 Hz hum	Move away from chargers and power strips. Never record while charging.
Signal jumps when moving	Cable movement. Keep leads short and the device fixed to the cap.

06 RELATED DOCUMENTS

IB8-DS-001 Product Datasheet	**IB8-MD-002** Mechanical & Assembly Drawing	**IB8-PCB-003** PCB & Manufacturing Specification
IB8-PIN-004 Connector, Electrode & Battery Guide	**IB8-SW-006** Bluetooth Interface & Software Guide	

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